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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE B

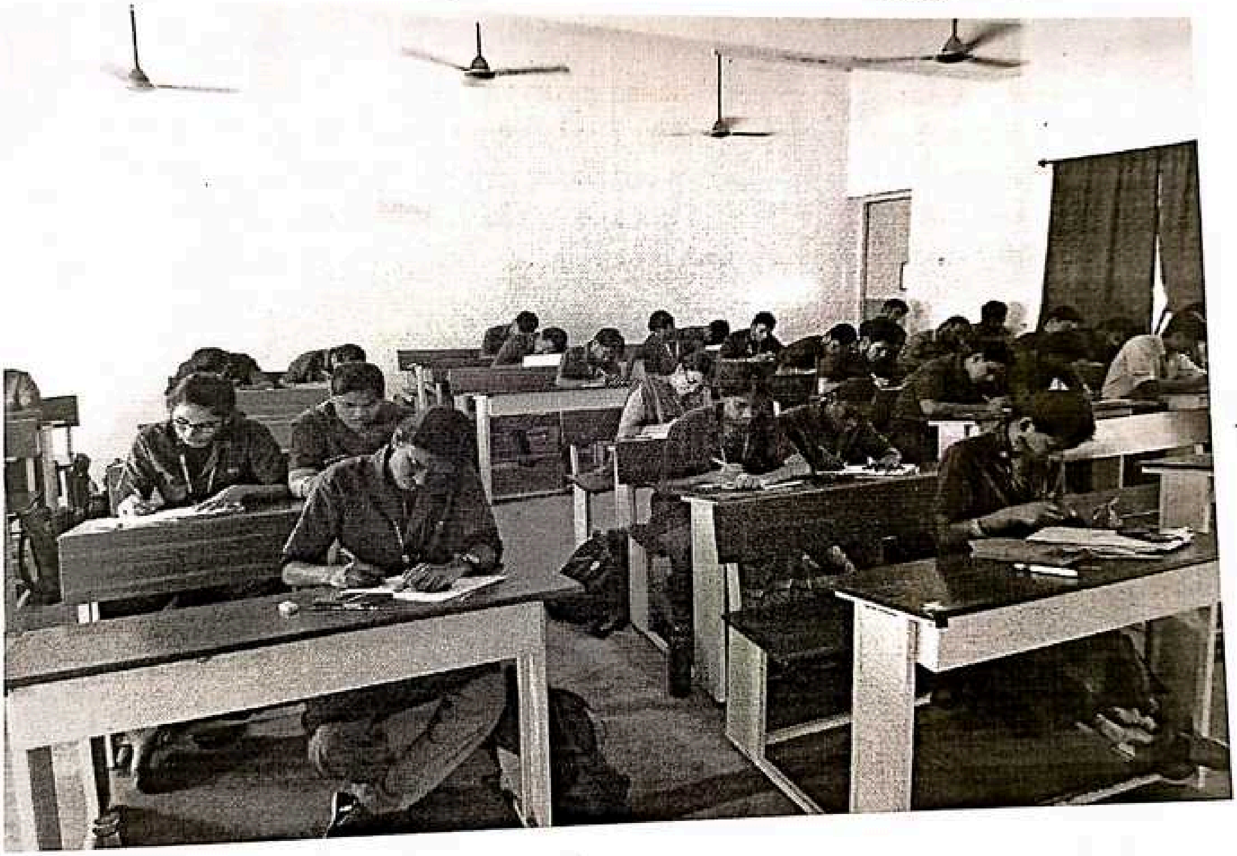
Course Code & Title: EC8552 Computer Architecture and Organization

Name of the Faculty member (s): Dr.R.Rajalakshmi

Active Learning Description

- **Unit / Topic:** Unit-I / Basics of a Computer system:Ideas
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO1
- **Activity Chosen:** One minute paper

Date: 13.08.2022



13/8/22
Signature of Faculty Member

13/8/22
HoD

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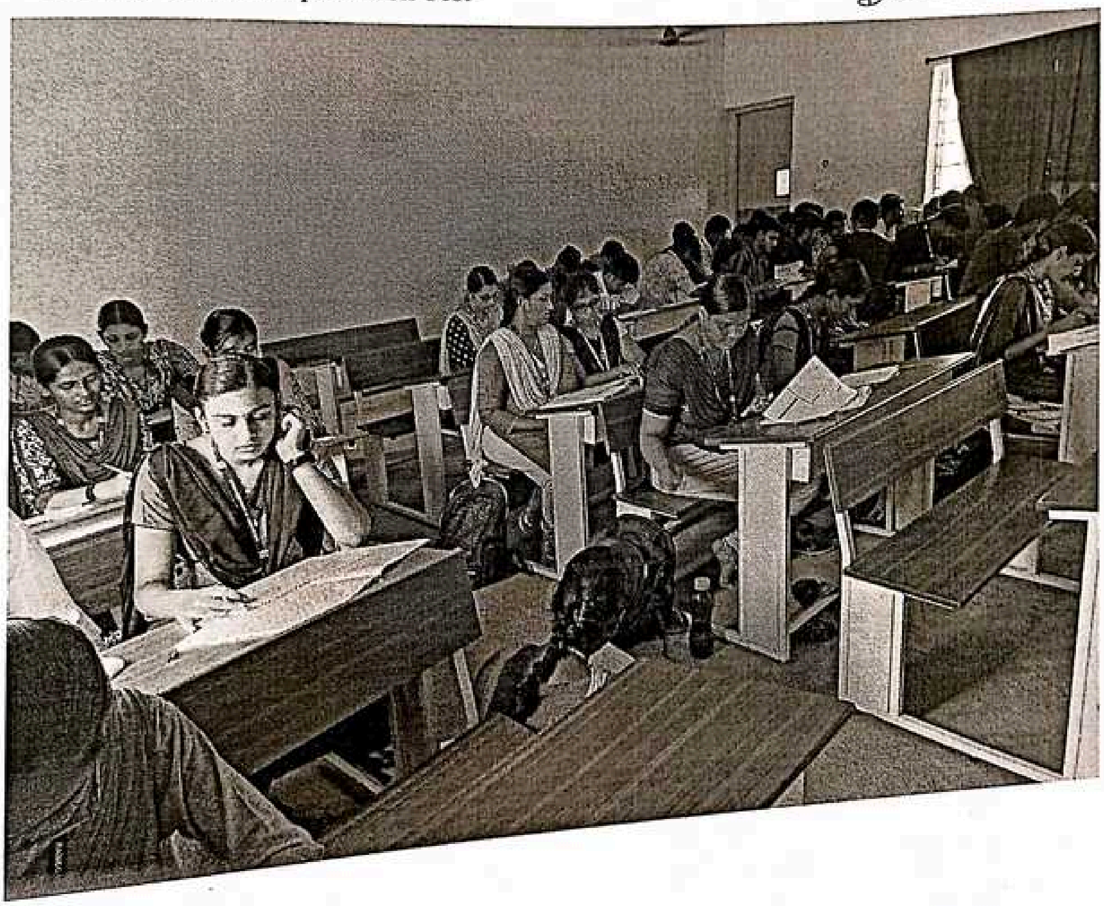
Active Learning Description

- **Unit / Topic: Unit-II /** Floating Point Arithmetic
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Open Book Test

Date: 17.9.2022



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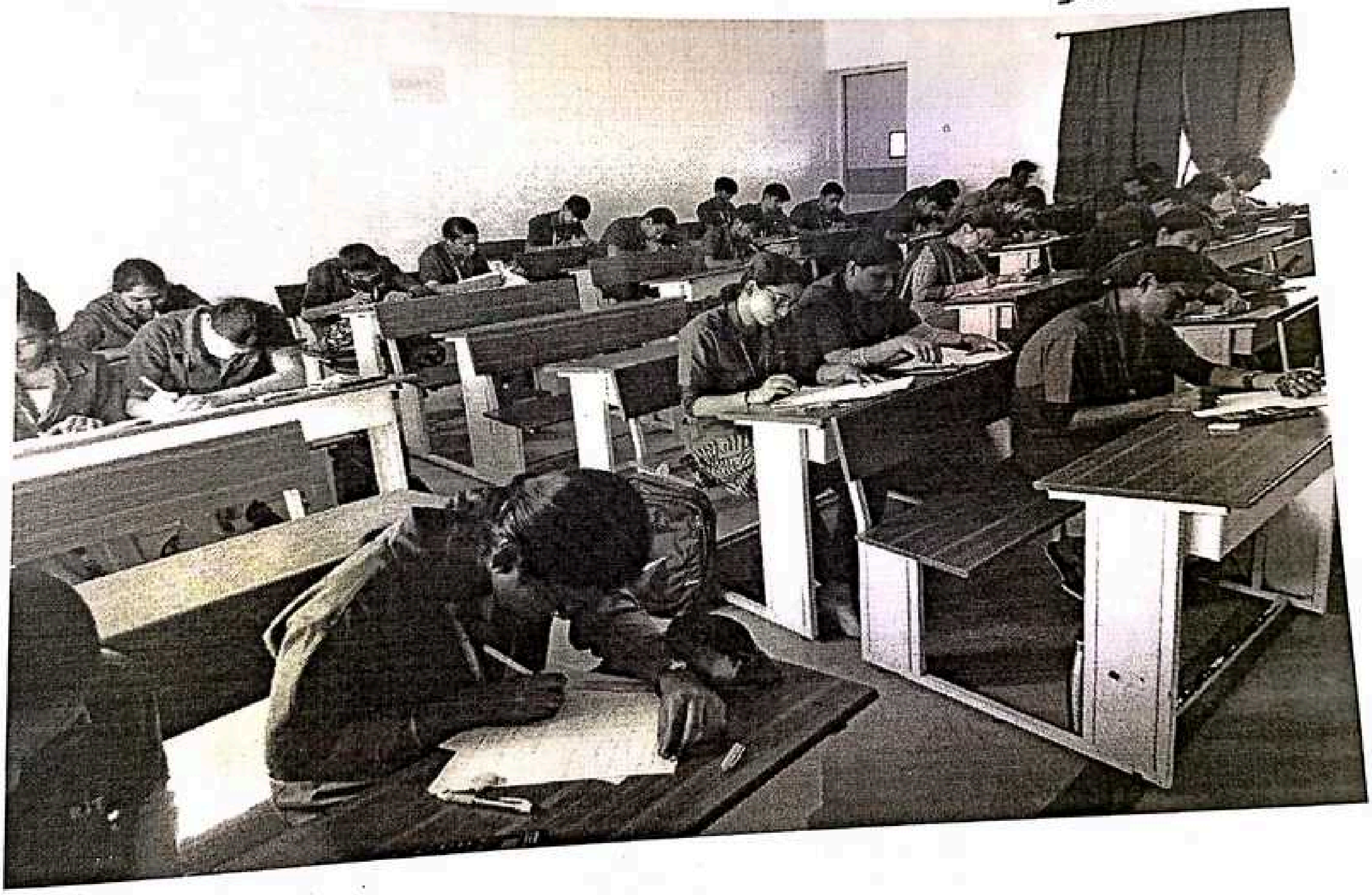
Active Learning Description

- **Unit / Topic: Unit-III /** Data Hazards: Forwarding versus Stalling
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** Exit Slips

Date : 30.9.22



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Active Learning Description

- **Unit / Topic: Unit-III /** Parallelism via Instructions
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TL09
- **Activity Chosen:** Crossword puzzle

Date: 6.10.22

RRajalakshmi 6/10/22
Signature of Faculty Member

6/10/22
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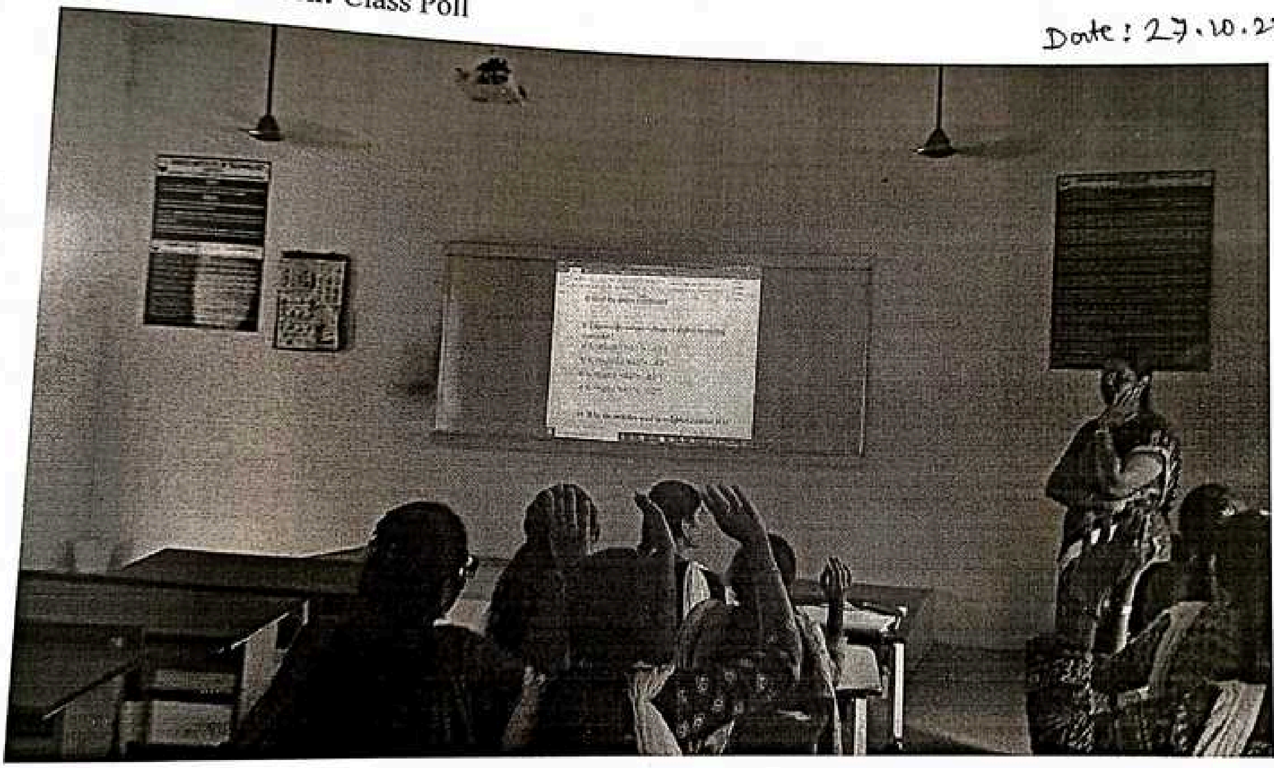
Active Learning Description

- **Unit / Topic:** Unit-IV / Internal Communication Methodologies
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TL13
- **Activity Chosen:** Class Poll

Date : 27.10.22



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Name of the Faculty member (s): Dr.R.Rajalakshmi

Innovative practice

Innovative Practice Description

- **Unit / Topic:** Unit 5 – Parallel processing architectures and challenges
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TL14
- **Activity Chosen:** Flipped Class
- **Justification:**

Parallel processing architectures and challenges topic were chosen for the flipped class. Advanced Computer Architecture are taught by Unit 3. The topic of " Parallel processing architectures and challenges" was chosen for the flipped class because the students were already familiar with Computer Organization and instructions as well as memory organization. Simple logic can be used to learn this subject on your own. The pupils have the opportunity to explore the subject to learn how the operations are changed. This flipped class exercise was chosen as a result to get them thinking about Parallel processing architectures and challenges and its uses.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

Students were given an outline of the flipped class activity before it began. Each group of students received a topic from the whole set that they had requested after being divided into smaller groups. The pupils were given data and internet resources about their subject. The time allotted for presentation preparation is ten minutes. The audience will listen as each group presents the topic they have chosen. There will be a question-and-answer period after the presentation so that attendees can share their opinions and concerns. This allowed all groups to communicate the ideas they had gathered about their respective subjects in class.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO11	PSO1	PSO2	PSO3
CO5	3	2	2	2	1	1	2	1

- **Images / Screenshot of the practice:**


- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    The students' exposure to various viewpoints during group discussions and their ability to apply theory to actual problems helped them develop their presentation skills. They admired the innovative thought that went into the work, which may be achieved through this sort of endeavor. They might receive ideas for the presentation from educational resources, which would help them put it together.
  - ***Benefit of the practice:***
  - Students successfully completed this task. They theoretically planned the presentation and set out the task for their chosen topic in a simple-to-understand manner. Everyone in the group understood the topics well. Students who worked in groups were better able to comprehend subjects because they were allowed to share thoughts. Each group member was skilled at explaining the concepts of various parallel processing Architectures and how they may be used.
  - ***Challenges faced in implementation:***
    Some members created the presentation material that lacked necessary information.
    Some students found it difficult to deliver due to insufficient communication.
    Some pupils used more time than was permitted.
    It was difficult for some pupils to address the concerns/questions voiced by other team members.

TEXT BOOKS:

1. David A. Patterson and John L. Hennessey, —Computer Organization and Designl, Fifth edition, Morgan Kauffman / Elsevier, 2014. (UNIT I-V)
2. Miles J. Murdocca and Vincent P. Heuring, —Computer Architecture and Organization: An Integrated approachl, Second edition, Wiley India Pvt Ltd, 2015 (UNIT IV,V)

REFERENCES

1. V. Carl Hamacher, Zvonko G. Varanesic and Safat G. Zaky, —Computer Organization—, Fifth Edition, Mc Graw-Hill Education India Pvt Ltd, 2014.
2. William Stallings —Computer Organization and Architecturel, Seventh Edition, Pearson education,2006.
3. Govindarajalu, —Computer Architecture and Organization, Design Principles and Applications",Second edition, McGraw-Hill Education India Pvt Ltd, 2014.

5/11/22

Signature of Faculty Member

5/11/22

HOD